



Bias Reduction and Fairness in Clustering

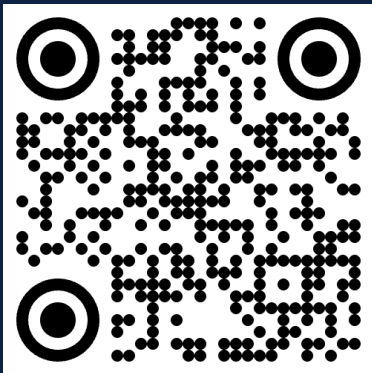
The Advanced R Series: Clustering & Classification

Slides and code created by Jesse Ghashti

March 18, 2026

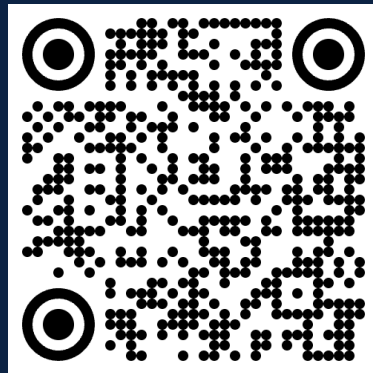
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GitHub code and
slides for today's
workshop (and pre-
vious workshops)



Alternatively, code/slides available at the bottom of
<https://csc-ubc-okanagan.github.io/workshops/>

Workshop Series Overview



Session	Topic	Date/Time
1	Hierarchical Models	Jan 14, 12:00 PM
2	Centroid-Based Models	Jan 19, 3:30 PM
3	Fuzzy Clustering	Jan 28, 1:00 PM
4	Distribution-Based Models	Feb 2, 3:30 PM
5	Density-Based Models	Feb 9, 3:30 PM
6	Graph-Based Models	Feb 23, 3:30 PM
7	Mixed-Type Data Quantification	Mar 4, 1:00 PM
8	Mixed-Type Data Clustering	Mar 11, 1:00 PM
9	Bias Reduction and Fairness	Mar 18, 1:00 PM
10	Dimensionality Reduction 1 of 2	Mar 23, 3:30 PM
11	Dimensionality Reduction 2 of 2	Mar 30, 3:30 PM

What we learned...

- Applied Gower's distance for mixed-type data clustering
- Learned PAM (Partitioning Around Medoids) — uses actual observations as centres
- Introduced K -Prototypes — extends K -means with mode-based categorical matching
- Used MDS to visualize dissimilarity matrices
- Applied spectral clustering with Gaussian kernel similarity
- Compared methods on simulated workforce data and heart disease records

Today we will...

- Motivate why fairness matters in clustering
- Define group fairness metrics: balance, disparate impact, composition deviation
- Define individual fairness (Kleindessner et al., 2020)
- Learn the **fairlet decomposition** algorithm (Chierichetti et al., 2017)
- Apply **linear residualization** as a preprocessing approach
- Compare methods on real COMPAS recidivism data

Today we require...

```
library('cluster')      # daisy, pam, silhouette
library('ggplot2')      # visualization
library('dplyr')        # data manipulation
library('tidyr')        # pivot_wider
library('gridExtra')    # plot arrangement
library('viridis')      # color palettes
```

Why Fairness in Clustering?



Clustering is used for consequential decisions

- Credit scoring: Grouping applicants into risk categories
- Healthcare: Allocating resources based on patient segments
- Criminal justice: Risk assessment for bail/sentencing decisions
- Hiring: Segmenting job applicants for screening
- Marketing: Targeting groups for products/services

If clusters correlate with protected attributes—such as race, sex, age, etc.—downstream decisions may be discriminatory (even if the algorithm never explicitly uses those attributes).

So far we've optimized for cluster quality. Today we wonder if are our clusters *fair*?

COMPAS: Correctional Offender Management Profiling for Alternative Sanctions

- Previously used by US courts to assess offenders likelihood of reoffending
- Produces risk scores for violence, recidivism, failure to appear
- A ProPublica investigation in 2016 found racial bias in predictions
 - Black defendants more likely to be falsely flagged as high-risk
 - White defendants more likely to be falsely flagged as low-risk

Today we'll cluster COMPAS data and examine whether standard clustering methods produce balanced clusters with respect to ethnicity — and what we can do if they don't.

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

DISCLAIMER 1 — This workshop uses COMPAS data solely for illustration of fair clustering methods. We simplified the ethnicity to a binary variable, subsampled for computational considerations, and treated risk scores as ground truth variables. These choices would be inappropriate for ANY real policy analysis. The fairlet implementation (Method 1) we describe is a simplified algorithm, not the optimal minimum-cost flow approach from the original paper. Linear residualization (Method 2) is a basic preprocessing technique that only removes linear associations. The methods shown are introductory; a true fairness pipeline in practice requires careful consideration of what fairness means in context of the data and problem being solved, domain expertise, legal review, and ongoing auditing, etc.

NOTE ON TERMINOLOGY — We use “Black / Non-Black” rather than the original dataset’s “African-American” label. This binary grouping is a simplification required by the one of the presented algorithms, which is designed for two groups, not a statement about racial categories. Collapsing Caucasian, Hispanic, Asian, and other ethnicities into “Non-Black” obscures important heterogeneity and would be inappropriate for real fairness consideration. We use “Black” as the broader term used more in contemporary usage, while acknowledging that neither label captures the full complexity of racial identity.

```
compasUrl <- "https://raw.githubusercontent.com/propublica/compas-analysis/master/compas-scores-raw.csv"
compasRaw <- read.csv(compasUrl)

riskLong <- compasRaw %>%
  filter(DisplayText %in% c("Risk of Violence", "Risk of Recidivism", "Risk of Failure to Appear")) %>%
  mutate(scale = case_when(DisplayText == "Risk of Violence" ~ "violence",
    DisplayText == "Risk of Recidivism" ~ "recidivism", DisplayText == "Risk of Failure to Appear" ~ "fta"))
scoresWide <- riskLong %>% select(AssessmentID, scale, RawScore) %>%
  pivot_wider(names_from = scale, values_from = RawScore, names_prefix = "raw_")

compas <- compas %>% mutate(EthBinary = ifelse(
  Ethnicity == "African-American", "Black", "Non-Black"), EthBinary = factor(EthBinary))
compas <- compas %>% group_by(EthBinary) %>% slice_sample(n = 500) %>% ungroup()
```

Features for clustering

raw_violence: Violence risk score

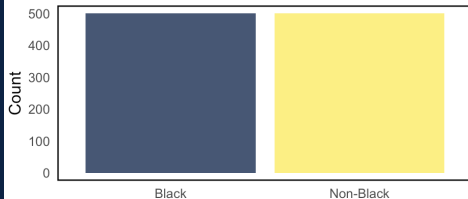
raw_recidivism: Recidivism risk score

raw_fta: Failure to appear risk score

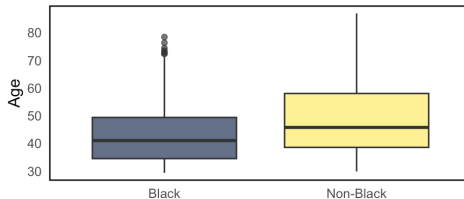
Age: Defendant age

Protected attribute: Ethnicity

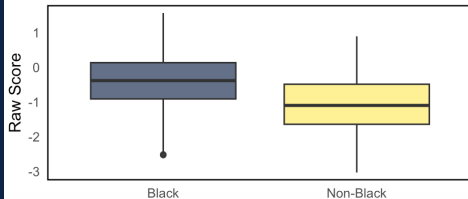
Binary Ethnicity Distribution



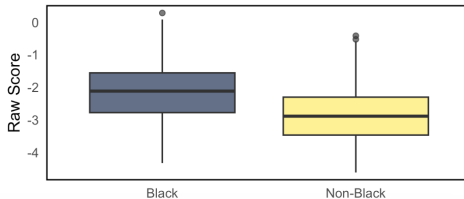
Age by Ethnicity



Recidivism Score by Ethnicity



Violence Score by Ethnicity



Ethnicity	n	Mean Age	Mean Recid.	Mean Viol.	Mean FTA
Black	500	43.6	-0.44	-2.15	19.3
Non-Black	500	48.8	-1.08	-2.84	17.8

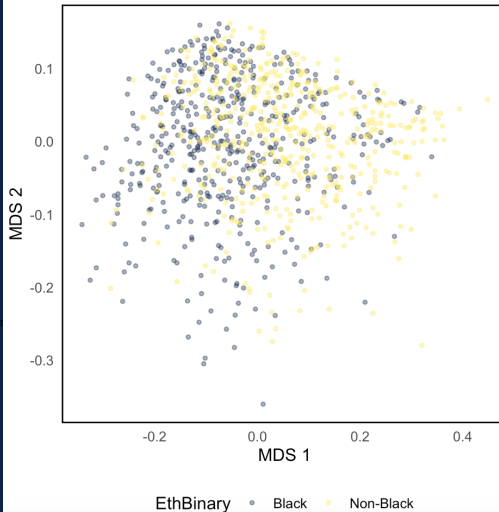
NOTE that risk scores differ between ethnic groups. If we cluster on these scores, will the clusters reflect this disparity?

Baseline Clustering with PAM and K -Means

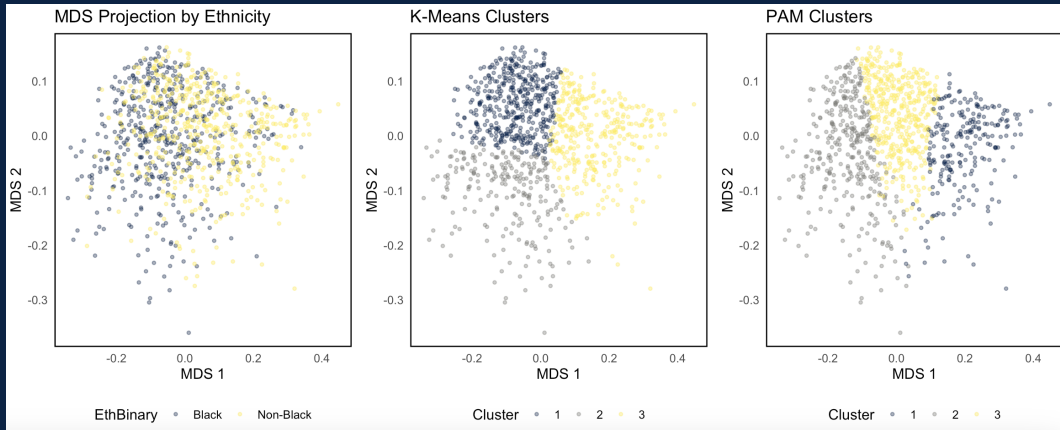


```
compasForClust <- compas %>%  
  select(raw_violence, raw_recidivism, raw_fta, Age)  
  
compasScaled <- scale(compasForClust)  
gowerCompas <- as.matrix(daisy(compasForClust, metric = "gower"))  
  
mdsCompas <- cmdscale(gowerCompas, k = 2)  
compas$mds1 <- mdsCompas[, 1]  
compas$mds2 <- mdsCompas[, 2]  
  
# K-means baseline  
set.seed(2026)  
kmeansBaseline <- kmeans(compasScaled, centers = 3, nstart = 25, iter.max = 100)  
compas$clusterKmeans <- factor(kmeansBaseline$cluster)  
  
# PAM baseline  
pamBaseline <- pam(gowerCompas, k = 3, diss = TRUE)  
compas$clusterPAM <- factor(pamBaseline$clustering)
```

MDS Projection by Ethnicity



Baseline Results



Group Fairness: Clusters should have balanced group proportions

Balance (Chierichetti et al., 2017): For a cluster C with r points from group Red and b points from group Blue:

$$\text{balance}(C) = \frac{\min(r, b)}{\max(r, b)} \in [0, 1]$$

- Balance = 1: Perfect parity (equal representation)
- Balance = 0: One group completely absent
- For a clustering, report **min** balance across all clusters (worst case)

A clustering is called **t -balanced** if every cluster has balance $\geq t$.

Disparate Impact & Composition Deviation

Disparate Impact Ratio: For cluster k , ratio of assignment rates between groups:

$$DI_k = \frac{P(\text{cluster} = k \mid \text{minority})}{P(\text{cluster} = k \mid \text{majority})}$$

- $DI \approx 1$: Equal assignment rates
- $DI < 0.8$: Often considered evidence of disparate impact (“80% rule”)

Composition Deviation: Average absolute difference from population proportions:

$$\text{CompDev} = \frac{1}{K} \sum_{k=1}^K \sum_g |p_{kg} - \pi_g|$$

where p_{kg} is proportion of group g in cluster k and π_g is overall proportion.

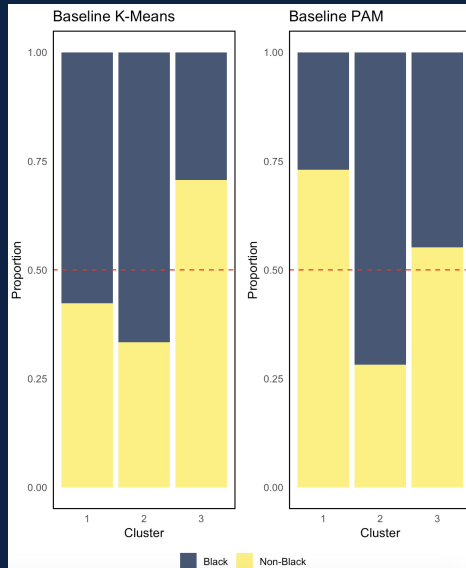

```
computeBalance <- function(clusters, sensitive) {  
  tab <- table(clusters, sensitive)  
  clusterBalances <- apply(tab, 1, function(row) {  
    if (max(row) == 0) return(0)  
    min(row) / max(row)  
  })  
  return(list(minBalance = min(clusterBalances),  
             perCluster = clusterBalances,  
             avgBalance = mean(clusterBalances)))  
}  
  
computeDisparateImpact <- function(clusters, sensitive) {  
  tab <- table(clusters, sensitive)  
  rates <- prop.table(tab, margin = 2)  
  clusterDIs <- apply(rates, 1, function(row) min(row) / max(row))  
  min(clusterDIs)  
}  
  
computeCompositionDeviation <- function(clusters, sensitive) {  
  overallProp <- prop.table(table(sensitive))  
  tab <- table(clusters, sensitive)  
  clusterProps <- prop.table(tab, margin = 1)  
  totalDev <- 0  
  for (k in 1:nrow(clusterProps)) {  
    totalDev <- totalDev + sum(abs(clusterProps[k, ] - overallProp))  
  }  
  totalDev / nrow(clusterProps)  
}
```

Baseline Fairness Evaluation



Baseline Fairness Scores

Metric	<i>K</i> -Means	PAM
Min Balance	0.416	0.369
Avg Balance	0.550	0.525
Comp. Deviation	0.300	0.333
Disp. Impact	0.416	0.369



But what are the clusters?



Baseline K -Means Cluster Centroids

Cluster	Violence	Recidivism	FTA	Age
1	-2.02	-0.46	16.2	36.8
2	-1.96	-0.00	25.8	48.3
3	-3.45	-1.63	16.6	56.4

Baseline PAM Cluster Medoid Profiles

Cluster	Violence	Recidivism	FTA	Age
1	-3.65	-1.75	17.0	61.8
2	-1.61	0.10	21.9	40.7
3	-2.59	-0.92	16.6	42.2

Individual Fairness (Kleindessner et al., 2020)

Group fairness: Clusters have balanced proportions of protected groups.

Individual fairness: Similar individuals should be treated similarly.

One notion of individual fairness states a point x_i is individually fair if, on average, it is closer to points in its own cluster than to points in any other cluster. Formally,

$$\frac{1}{|C_{x_i}| - 1} \sum_{j \in C_{x_i}, j \neq i} d(x_i, x_j) \leq \frac{1}{|C_k|} \sum_{j \in C_k} d(x_i, x_j) \quad \forall k \neq C_{x_i}$$

Individual fairness asks “Is each person in the cluster where they belong based on similarity?” — regardless of their protected attribute.

```
computeIndividualFairness <- function(distMat, clusters) {  
  distMat <- as.matrix(distMat)  
  n <- nrow(distMat)  
  violations <- 0  
  
  for (i in 1:n) {  
    ownCluster <- clusters[i]  
    ownMembers <- which(clusters == ownCluster & (1:n) != i)  
    if (length(ownMembers) == 0) next  
  
    avgDistOwn <- mean(distMat[i, ownMembers])  
  
    otherClusters <- setdiff(unique(clusters), ownCluster)  
    for (k in otherClusters) {  
      kMembers <- which(clusters == k)  
      if (length(kMembers) > 0) {  
        avgDistK <- mean(distMat[i, kMembers])  
        if (avgDistOwn > avgDistK) { violations <- violations + 1; break }  
      }  
    }  
  }  
  return(violations / n)  
}
```

The First Paper in Fair Clustering

Idea: Decompose data into small, balanced “fairlets” first, then cluster the fairlets.

A *fairlet* is a minimal subset that satisfies fairness constraints.

Algorithm

1. Match points from different groups to form balanced pairs
2. Compute a representative (centroid) for each fairlet
3. Run standard clustering on fairlet representatives
4. Assign each point to the cluster of its fairlet representative

For our sampled data containing 500 observations from each group, a balance of 1 (perfect parity) is guaranteed when fairlets are simply pairs of one point from each group.

Fairlet Decomposition Intuition



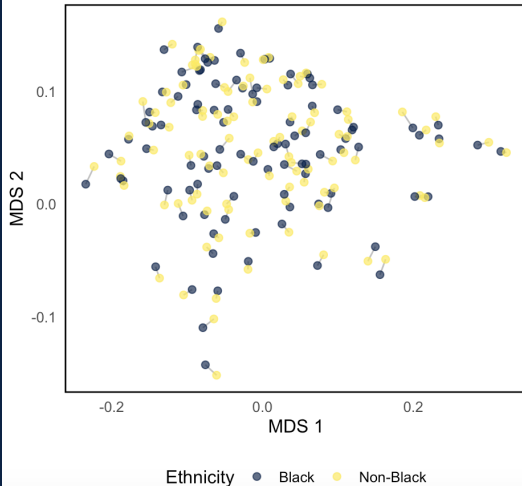
Without fairlets:

Clustering may group similar points together, but similar points from one ethnicity may cluster separately from similar points of another.

With fairlets:

We first pair similar points from *different* groups. These pairs become atomic units. Any clustering of these units is guaranteed to be balanced.

Select Fairlet Decompositions



```
buildFairlets <- function(features, sensitive, distMat = NULL) {  
  groups <- levels(factor(sensitive))  
  idx0 <- which(sensitive == groups[1])  
  idx1 <- which(sensitive == groups[2])  
  nPairs <- min(length(idx0), length(idx1))  
  
  if (is.null(distMat)) distMat <- as.matrix(dist(features))  
  else distMat <- as.matrix(distMat)  
  
  fairlets <- list()  
  used0 <- rep(FALSE, length(idx0)); used1 <- rep(FALSE, length(idx1))  
  
  for (i in 1:nPairs) {  
    bestDist <- Inf; best0 <- NA; best1 <- NA  
    for (j in which(!used0)) {  
      for (k in which(!used1)) {  
        d <- distMat[idx0[j], idx1[k]]  
        if (d < bestDist) { bestDist <- d; best0 <- j; best1 <- k }  
      }  
    }  
    used0[best0] <- TRUE; used1[best1] <- TRUE  
    fairlets[[i]] <- c(idx0[best0], idx1[best1])  
  }  
  return(fairlets)  
}
```


Clustering on Fairlet Centroids



```
#### SEE FULL R CODE SCRIPT FOR ALL FUNCTIONS ####
```

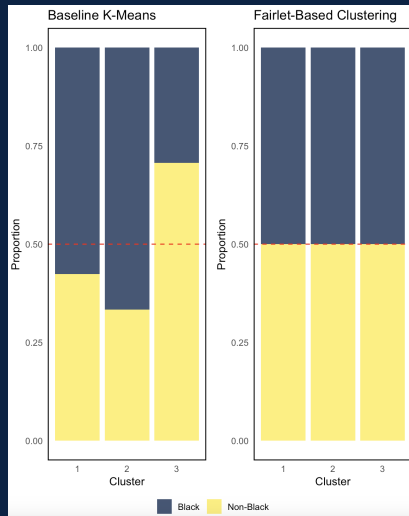
```
fairlets <- buildFairlets(compasScaled,  
                          compas$EthBinary,  
                          gowerCompas)
```

```
fairletCentroids <- computeFairletCentroids(  
  compasScaled, fairlets)
```

```
set.seed(2026)
```

```
kmeansFairlet <- kmeans(fairletCentroids,  
                        centers = 3,  
                        nstart = 25)
```

```
compas$clusterFairlet <- factor(  
  assignFairletClusters(fairlets,  
                        kmeansFairlet$cluster,  
                        nrow(compas)))
```



Fairlet	Method	Fairness
Metric	<i>K</i> -Means	Fairlet
Min Balance	0.416	1.000
Avg Balance	0.550	1.000
Comp. Dev.	0.300	0.000
Disp. Impact	0.416	1.000

Fairlets achieve our group fairness definitions by construction — every cluster inherits the balance of the fairlets it contains.

Baseline *K*-Means Cluster Centroids

Cluster	Violence	Recidivism	FTA	Age
1	-2.02	-0.46	16.2	36.8
2	-1.96	-0.00	25.8	48.3
3	-3.45	-1.63	16.6	56.4

Fairlets Cluster Centroids

Cluster	Violence	Recidivism	FTA	Age
1	-3.38	-1.60	15.3	54.2
2	-2.51	-0.65	23.3	52.8
3	-2.15	-0.59	16.1	38.1

Preprocessing Approach: Remove Protected Attribute First Moment Effect

Idea: Regress each continuous feature on the protected attribute and use the residuals for clustering.

$$X_j^{\text{resid}} = X_j - \hat{\mathbb{E}}[X_j \mid A]$$

where A is the protected attribute

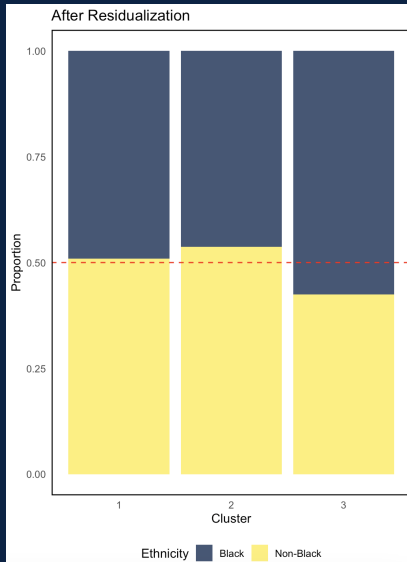
Removes linear association between features and protected attribute. The residualized features have the same mean across groups.

Limitation is clearly that it only removes *linear* effects — nonlinear associations may still be present. This does not account for causal effects or interactions

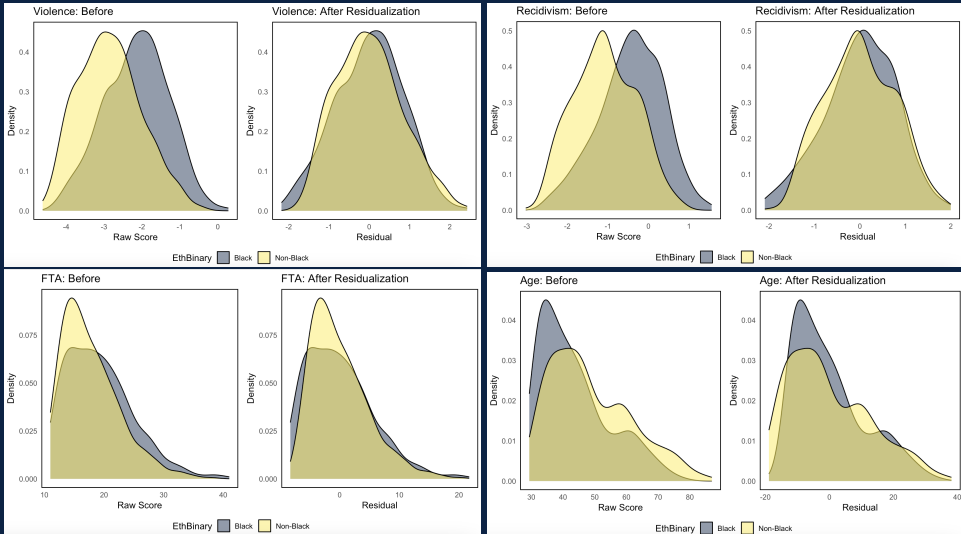
Applying Residualization



```
continuousVars <- c("raw_violence",  
                    "raw_recidivism",  
                    "raw_fta", "Age")  
  
compasResid <- compas  
  
for (var in continuousVars) {  
  formula <- as.formula(  
    paste(var, "~ EthBinary")  
  )  
  model <- lm(formula, data = compas)  
  compasResid[[var]] <- residuals(model)  
}  
  
compasResidScaled <- scale(  
  compasResid[, continuousVars])  
  
set.seed(2026)  
kmeansResid <- kmeans(compasResidScaled,  
  centers = 3,  
  nstart = 25)  
compas$clusterResid <- factor(  
  kmeansResid$cluster)
```



Effect of Residualization on Distributions



Method Comparison: Fairness vs Quality

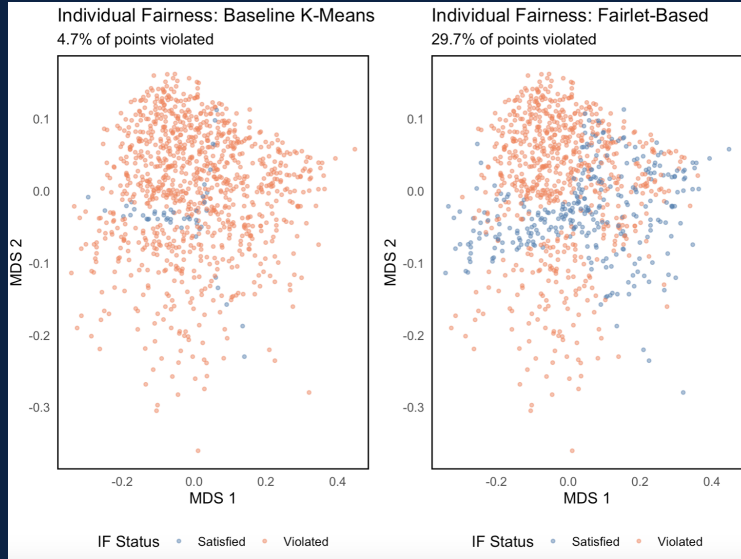


Method	Min Bal.	Avg Bal.	Comp. Dev.	IF Viol.	Silhouette
Baseline K -Means	0.416	0.550	0.300	0.047	0.345
Fairlets	1.000	1.000	0.000	0.297	0.165
Residualization	0.738	0.855	0.081	0.134	0.340

Tradeoff: Improving fairness (higher balance, lower composition deviation, etc.) may reduce cluster quality (lower silhouette). This is called the “price of fairness.”

Tradeoff: Improving group fairness may impact individual fairness definitions negatively as a result. It is argued these definitions are in direct opposition of one another.

Individual Fairness Violations



Group Fairness

- Clusters have balanced proportions
- Protects groups as a whole
- Measured by: balance, disparate impact, composition deviation
- May place dissimilar individuals together

Individual Fairness

- Similar individuals treated similarly
- Protects individuals
- Measured by: avg distance to own vs other clusters
- May still produce imbalanced clusters

These two notions can conflict. A clustering that is group-fair may violate individual fairness, and vice versa. The choice depends on the application context.

- Standard clustering can produce ethnically imbalanced clusters even without using ethnicity as a feature
- Balance measures the ratio of group sizes within each cluster
- Fairlets decompose data into balanced pairs before clustering — guarantees group fairness
- Residualization removes linear effects of protected attributes
- Individual fairness asks whether each point is closer to its own cluster
- There is typically a tradeoff between fairness and cluster quality
- The appropriate fairness notion depends on the application context

Additional Questions?
Book an Appointment!



Next Workshop: Mar 23, 3:30 PM

Dimensionality Reduction 1 of 2

-> High-dimensional data is hard to cluster and visualize.

-> Next workshop introduces PCA and related techniques for reducing dimensionality while preserving structure.

-> We'll see how dimensionality reduction can improve clustering performance.

Thank You!

Questions?

Workshop Materials:

<https://csc-ubc-okanagan.github.io/workshops/>

Contact:

jesse.ghashti@ubc.ca